

Limit Order Book Execution Plan

Algo Type1 Signals and Internal Flow Optimization

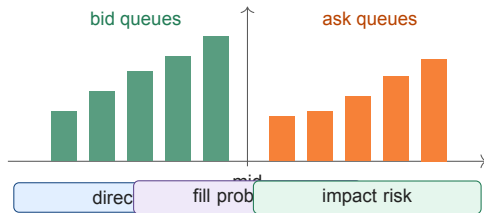
15-minute leadership briefing | June 4, 2026

■ One-Page Context

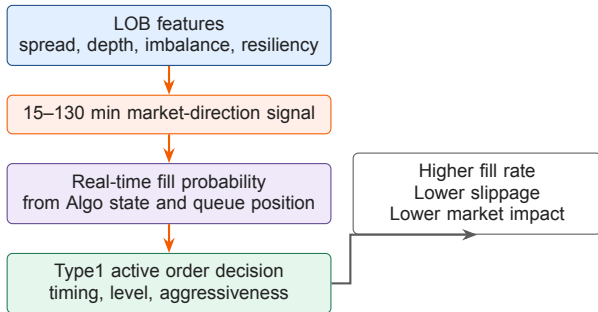
Why LOB matters

The order book is the closest observable state before execution: price direction, queue depth, fill probability, and impact risk are visible together.

- ▶ Keep the technical introduction to one page.
- ▶ Use LOB features to improve execution, not only to predict price.
- ▶ Focus on two deliverables: **Algo Type1** and **department-level internal flow**.



■ Output 1: Algo Type1 Execution Alpha



Business use

Support the current Algo instance for **Type1 active order placement**.

- ▶ Signal horizon: 15, 30, 60, 90, 130 minutes.
- ▶ Combine market direction with execution probability.
- ▶ Improve fill probability when direction is favorable.
- ▶ Avoid chasing in adverse LOB regimes.

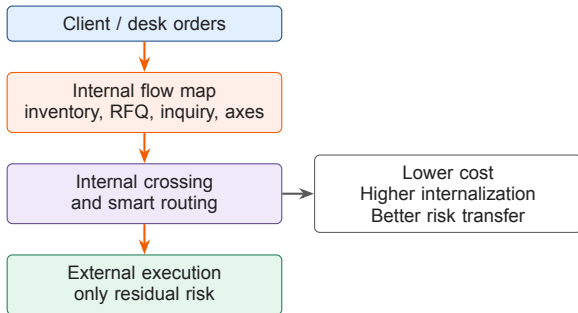
■ Output 1: Delivery Plan and KPIs

Prototype path

1. Build LOB feature stream and direction labels.
2. Train interpretable baseline signals by horizon.
3. Join signal with Algo fill-probability engine.
4. Run shadow trading, then controlled live A/B test.

KPI	Measurement
Fill rate uplift	Type1 fills vs. baseline by regime
Slippage saving	bps vs. arrival / decision price
Impact control	post-trade adverse move
Model stability	hit rate and IC across horizons
●	●
Data	Shadow
	A/B
	Scale

■ Output 2: Department Internal Flow Engine



Department-level target

Develop an internal flow optimization layer that reduces transaction cost through internal crossing, better routing, and pre-trade liquidity selection.

- ▶ Convert historical flow into actionable liquidity intelligence.
- ▶ Match compatible internal demand before external trading.
- ▶ Use LOB and flow signals to choose execution timing.

■ Value Case: P&L Math and Sensitivity

Fixed-income approximation

$$\Delta \text{PnL} \approx \text{Notional} \times \text{Duration} \times \text{Saving}_{bp} \times 10^{-4}$$

- ▶ 2025 volume: **RMB 45tn.**
- ▶ Average duration assumption: 2 years.
- ▶ At 0.01 bp yield saving:
 $45tn \times 2 \times 0.01 \times 10^{-4} = \text{RMB 90mm}.$
- ▶ RMB 1bn requires about **0.11 bp** yield saving under the same assumptions.

Yield saving	Approx. annual P&L
0.01 bp	RMB 0.09 bn
0.05 bp	RMB 0.45 bn
0.10 bp	RMB 0.90 bn
0.11 bp	RMB 0.99 bn
0.20 bp	RMB 1.80 bn

Message

The value case remains material, but the RMB 1bn claim should be framed as a **0.10–0.11 bp** saving target, not 0.01 bp.

■ Roadmap and Leadership Ask

Two workstreams

- ▶ **Algo Type1:** 15–130 min direction signal plus fill-probability integration.
- ▶ **Internal flow:** internal crossing, routing, and liquidity selection engine.

Near-term milestones

- ▶ 2 weeks: data audit, baseline features, signal dashboard.
- ▶ 4–6 weeks: shadow trading and internal-flow simulator.
- ▶ 8–12 weeks: controlled production pilot.

Decision needed

- ▶ Confirm priority instruments and Algo instance.
- ▶ Approve access to LOB, order, fill, and internal flow data.
- ▶ Align on P&L attribution: slippage, market impact, and internal crossing savings.

**Deliver execution alpha first,
then scale it into a
department-level flow product.**